# Andrés Esteban Merino Toapanta

Mathematician
Full professor


October 18, 1990
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Ecuadorian


## Online Profiles

Orcid
Github
LinkedIn
linktr.ee/aemerinot

## Languages

- Español ●●●●●
- English ●●●●○
- Française ●●○○○

## Areas of Interest

- Descriptive Set Theory
- Foundations of Mathematics
- Data Science
- Machine Learning
- Editing and Composition in LaTeX
- Mathematics Education

## Education

feb-2024 **Master's Degree in Data Science** Universitat Oberta de Catalunya

Thesis: Automatic Image Recognition for Plant-Pollinator Network Reconstruction

mar-2017 **Master's Degree in Pure and Applied Mathematics** Universidad Central del Ecuador

Thesis: Classification of Countable Compact Subsets in some Polish Spaces

oct-2014 **Master 1** Université Jean Monnet

Master Mathematiques 1

oct-2014 **Mathematician** Escuela Politécnica Nacional

Thesis: Classification of Countable Compact Subsets of the Reals
*Summa Cum Laude*

jul-2008 **High School Diploma** Colegio Municipal Experimental "Sebastián de Benalcázar"

Specialization in Physics and Mathematics

## Teaching Experience

apr-2017 **Full Professor** Pontificia Universidad Católica del Ecuador

Mathematics Professor in the Faculty of Exact and Natural Sciences (current full-time job: 7 years)

apr-2021 **Postgraduate Lecturer** Pontificia Universidad Católica del Ecuador

Lecturer in the Master's Degree in Experimental Sciences Education, major in Mathematics and Physics (part-time: 1 year and 6 months)

mar-2018 **Lecturer** Escuela Politécnica Nacional

Mathematics Professor in the Department of Basic Education (half-time: 12 months)

oct-2014 **Adjunct Professor** Escuela Politécnica Nacional

Mathematics Professor in the Faculty of Sciences (2 years and 5 months)

apr-2014 **Teacher** Colegio Municipal "Sebastián de Benalcazár"

Mathematics Teacher (4 months)

## Professional Experience (Non-teaching)

apr-2021 **Postgraduate Coordinator** Universidad Central del Ecuador

Coordinator of the Master's Degree in Pure and Applied Mathematics in the Faculty of Sciences (part-time: 1 year and 9 months)

jun-2019 **Editor** Ministerio de Educación

Editor of Mathematics modules for the 10th year of Basic General Education in the National Curriculum Directorate (1 month)

feb-2018 **Validator** Instituto Nacional de Evaluación Educativa (INEVAL)

Validation of item production 2018 in the Item Production and Safeguarding Directorate (2 months)

may-2015 **Monitor** Instituto de Altos Estudios Nacionales (IAEN)

Monitor of the virtual Mathematics course in the Continuing Education Center (CEC-EP) (1 month)

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## About Me

I have dedicated my career to teaching and research. In the latter, I collaborate in the areas of Descriptive Set Theory, Foundations of Mathematics, Mathematics Education, and Data Science. I have contributed knowledge and advancements through publications in scientific journals and participation in national and international conferences. My contributions include studies on the Cantor-Bendixson Derivative, Machine Learning models for data and image analysis, and innovative techniques for teaching Mathematics.

## Software

- LaTeX
- PSTricks
- GeoGebra
- Python
- Mathematica
- Inkscape
- R
- SQL
- Git
- TikZ

oct-2012 **Laboratory Assistant** Escuela Politécnica Nacional
Support in the seed research project "Flow over granular assemblies" (1 year)

dec-2011 **Editing Assistant** Abya Yala, Revista La Granja
Layout, diagramming, and template design (3 years)

mar-2011 **Laboratory Assistant** Escuela Politécnica Nacional
Editing in LaTeXand support in classes of Single Variable Calculus and Foundations of Mathematics (1 year and 3 months)

aug-2010 **Editing Assistant** Escuela Politécnica Nacional
Editing and publication of books and journals in the Faculty of Sciences Publications Unit (6 months)

nov-2010 **Editing Assistant** Analítika, revista de análisis estadístico
Editing, diagramming, and layout. (3 years)

## Appointments

mar-2025 **Head of the Service Unit for Curriculum Development and Innovation**
*Pontificia Universidad Católica del Ecuador, Faculty of Exact, Natural and Environmental Sciences*
Current appointment

jul-2023 **Coordinator of the School of Physical and Mathematical Sciences**
*Pontificia Universidad Católica del Ecuador, Faculty of Exact and Natural Sciences, School of Physical and Mathematical Sciences*
1 year and 8 months

mar-2020 **Coordinator of Virtual Classrooms**
*Pontificia Universidad Católica del Ecuador, Faculty of Exact and Natural Sciences, School of Physical and Mathematical Sciences*
3 years

apr-2021 **Coordinator of the Master's Degree in Pure and Applied Mathematics**
*Universidad Central del Ecuador, Faculty of Sciences*
1 year and 9 months

sep-2018 **Coordinator of the Vector Calculus Course**
*Escuela Politécnica Nacional, Department of Basic Education*
6 months

sep-2018 **Coordinator of the Linear Algebra Course**
*Escuela Politécnica Nacional, Department of Basic Education*
6 months

## University Courses Taught

- Machine learning
- Single Variable Calculus
- Vector Calculus
- Mathematical Analysis
- Mathematical Logic and Set Theory
- Real Analysis
- Foundations of Mathematics
- Ordinary Differential Equations
- Linear Algebra
- Fuzzy Logic
- Discrete Mathematics
- Mathematics for Graphic Design

## Courses and Conferences Delivered (last 5 years)

nov-2025 **Presentation: Exploring the Didactic Potential of ChatGPT Hallucinations in Teaching**
*VII Congreso Iternacional de Tecnología y Educación - TEDUCA 2025*
Universidad de Santander

jun-2025 **Conference: Exploring the Didactic Potential of ChatGPT Hallucinations in Teaching**
*Workshop Internacional INCOIN LEARNING: Inteligencia Artificial aplicada a la Educación Superior, edición PUCE 2025*
Pontificia Universidad Católica del Ecuador

aug-2024 **Conference: Integrating ChatGPT into the Flipped Classroom Methodology**
*Workshop INCOIN: Inteligencia Artificial en la Educación Superior, edición PUCE 2024*
Pontificia Universidad Católica del Ecuador

feb-2024 **Presentation: Use of ChatGPT for Teaching Differential Calculus**
*MEM 2024*
Universidad Antonio Nariño

oct-2023 **Presentation: Does ChatGPT Know How to Derive? An Experience in the Classroom**
*STEM Miami 2023 Seminar*
Broward International University

aug-2023 **Minicourse: Reinforcement Learning**
*Summer School in Machine Learning 2023*
Pontificia Universidad Católica del Ecuador (6 hours)

jul-2023 **Panel: Coordinators of Undergraduate and Postgraduate Programs in Data Science**
*I Meeting on Big Data and Artificial Intelligence Methods and Tools Applied to the Study of Social Phenomena in Ecuador*
Flacso Ecuador

sep-2022 **Presentation: Activity Guide for Mathematics Reinforcement Based on Gamification Directed at Upper Basic Level**
*II International Education Congress: Crisis, Creativity, and Educational Transformation*
Pontificia Universidad Católica del Ecuador

feb-2022 **Workshop: Innovative Didactics for Teaching and Learning Mathematics**
La Salle, Distrito Lasllista Norandino (30 hours)

nov-2021 **Video Presentation: Designing Fonts through Bézier Curves and GeoGebra**
*VIII International Workshop "Trends in Mathematics Education Based on Research in Alliance with the GeoGebra Latin American Community"*
Benemérita Universidad Autónoma de Puebla

sep- 2021 **Course: Intermediate Level GeoGebra Software Management**
Universidad Regional Amazónica Ikiam (10 hours)

## Conferences Attended (last 5 years)

nov-2025 **VII Congreso Iternacional de Tecnología y Educación - TEDUCA 2025**
*Universidad de Santander*
Santa Marta–Colombia (16 hours)

aug-2024 **INCOIN Workshop: Artificial Intelligence in Higher Education, PUCE 2024 edition**
*Pontificia Universidad Católica del Ecuador*
Quito–Ecuador (18 hours)

oct-2023 **STEM Miami Seminar 2023**
*Broward International University*
Miami–United States (8 hours)

sep-2022 **II International Education Congress: Crisis, Creativity, and Educational Transformation**
*Pontificia Universidad Católica del Ecuador*
Quito-Ecuador (40 hours)

jun-2022 **I International Symposium on Physics and Mathematics**
*Pontificia Universidad Católica del Ecuador*
Quito-Ecuador (10 hours)

feb-2022 **XII Symposium on Mathematics and Mathematics Education, XI International Congress on Mathematics Assisted by Computer (MEM2022)**
*Universidad Antonio Nariño*
Bogotá-Colombia (20 hours)

jan-2022 **Applied Data Science Research Congress – II National Congress of R Users Group**
*Escuela Politécnica Nacional*
Quito-Ecuador (34 hours)

nov-2021 **VIII International Workshop "Trends in Mathematics Education Based on Research in Alliance with the GeoGebra Latin American Community"**
*Benemérita Universidad Autónoma de Puebla*
Puebla-Mexico (10 hours)

## Teacher Training Courses (last 5 years)

jul-2025 **Challenge-Based Learning**
*Tecnológico de Monterrey*
Monterrey–México (26 hours)

sep-2025 **Diploma in University Teaching**
*Pontificia Universidad Católica del Ecuador*
Quito–Ecuador (140 hours)

feb-2025 **Advanced Exploration of Moodle Tools**
*Pontificia Universidad Católica del Ecuador*
Quito–Ecuador (40 hours)

jan-2025 **Designing AI-Mediated Learning Experiences**
*Tecnológico de Monterrey*
Monterrey–México (24 hours)

feb-2024 **AI to Improve the Teaching and Learning Process**
*IDEA-USFQ*
Quito-Ecuador (6 hours)

jul-2023 **Artificial Intelligence Applied to Education**
*Universidad Israel*
Quito-Ecuador (80 hours)

jul-2022 **PUCE INNOVA: Towards a University Pedagogical Transformation**
*Reimagine Education*
Barcelona-Spain (16 hours)

dec-2021 **Concept and Development of a Prototype for Educational Transformation Using the RIEDUSIS Methodology**
*Reimagine Education*
Barcelona-Spain (65 hours)

oct-2021 **Certification in Neuroscience and Education Applied to Teaching**
*Pontificia Universidad Católica del Ecuador*
Quito-Ecuador (50 hours)

apr-2021 **Creating Self-Evaluative Activities with GeoGebra**
*Instituto GeoGebra Extremeño*
Mérida-Spain (30 hours)

feb-2021 **Advanced Course for Virtual Classroom Management**
*Pontificia Universidad Católica del Ecuador*
Quito-Ecuador (24 hours)

## Other Training Courses (last 5 years)

ene-2026 **Application of topological data analysis**
*Sociedad Ecuatoriana de Estadística*
Ecuador (20 hours)

nov-2025 **Agentic AI**
*Actumlogos*
Mexico (36 hours)

sep-2025 **Digital Image Processing with Python and OpenCV**
*umaker*
Perú (40 hours)

mar-2025 **Introduction to Reinforcement Learning**
*CEDIA*
Cuenca–Ecuador (40 hours)

feb-2025 **Efficient Academy Writer with AI**
*Sociedad Ecuatoriana de Estadística*
Quito–Ecuador (20 hours)

feb-2025 **General Information Security Policy**
*Pontificia Universidad Católica del Ecuador*
Quito–Ecuador (20 hours)

nov-2024 **Elements of AI**
*University of Helsinki*
(50 hours)

nov-2024 **Gender-Based Violence Prevention**
*Pontificia Universidad Católica del Ecuador*
Quito–Ecuador (20 hours)

nov-2022 **Amazon Web Service for Data Science**
*SDC Learning*
Lima-Peru (64 hours)

mar-2021 **Machine Learning with Python**
*Universidad Peruana Cayetano Heredia*
Lima-Peru (32 hours)

## Publications

dec-2023 **Scientific Article**
*Merino, A. and Heredia, S. (2023). Relationship of the Cantor-Bendixson Derivative with Set Algebra. Selecciones Matemáticas, 10(2), 339-351.*
`https://doi.org/10.17268/sel.mat.2023.02.10`
Area: Descriptive Set Theory

jan-2023 **Scientific Article**
*Jiménez, S. and Merino, A. (2023). Machine Learning Models Based on CRISP-DM for the Analysis of Depression Levels in Students of the Escuela Politécnica Nacional. Latin-American Journal of Computing, 10(1), 22-43.*
`https://lajc.epn.edu.ec/index.php/LAJC/article/view/335`
Area: Data Science

aug-2022 **Scientific Article**
*Merino, A., Cueva, M., Sarmiento, M., and Paredes, A. (2022). Analysis of the Concepts of Variable and Constant in Students of the Unified General Baccalaureate of Ecuador. 593 Digital Publisher CEIT, 7(4-2), 175-185.*
`https://doi.org/10.33386/593dp.2022.4-2.970`
Area: Mathematics Education

jun-2021 **Scientific Article**
*Merino, A. and Ortiz-Castro, J. (2021). Continuity of Functions Based on Rearrangements of $\beta$-Expansions of a Number. Digital Journal of Mathematics, Education, and Internet, 22(1).*
https://doi.org/10.18845/rdmei.v22i1.5758
Area: Real Analysis

feb-2021 **Scientific Article**
*Álvarez-Samaniego, B. and Merino, A. (2021). Some Properties Related to the Cantor-Bendixson Derivative on a Polish Space. New Zealand Journal of Mathematics, 50, 207-218.*
http://dx.doi.org/10.17268/sel.mat.2020.02.04
Area: Descriptive Set Theory

dec-2020 **Scientific Article**
*Merino, A. and Heredia, S. (2020). A Generalization of the Cantor Ternary Set. Selecciones Matemáticas, 7(2), 222-233.*
http://dx.doi.org/10.17268/sel.mat.2020.02.04
Area: Real Analysis

may-2020 **Scientific Article**
*Merino, A. and Trujillo, J. C. (2020). Equivalence of the Axiom of Choice with the Function Redefinition Problem. Revista Politécnica, 45(2), 51-56.*
https://doi.org/10.33333/rp.vol45n2.05
Area: Foundations of Mathematics

nov-2019 **Scientific Article**
*Álvarez-Samaniego, B. and Merino, A. (2019). Countable Ordinal Spaces and Compact Countable Subsets of a Metric Space. Australian Journal of Mathematical Analysis and Applications, 16(2), article 12.*
Area: Descriptive Set Theory

jun-2019 **Scientific Article**
*Ortiz, J. and Merino, A. (2019). Schematic Representation of the Set Classification Problem. The Classification of the Quadratic Equation of Two Variables. Mathematics ESPOL-FCNM JOURNAL, 17(1), 1-14.*
Area: Mathematics Education

sep-2016 **Scientific Article**
*Álvarez-Samaniego, B. and Merino, A. (2019). A Primitive Associated to the Cantor-Bendixson Derivative on the Real Line. Journal of Mathematical Sciences: Advances and Applications 41 (1).*
http://dx.doi.org/10.18642/jmsaa_7100121692
Area: Descriptive Set Theory

junio 2012 **Booklet**
*Merino, A., Cueva, E. and Trujillo J.C. (2012). Single Variable Calculus: Summary and Solved Exercises. Publications Unit of the Faculty of Sciences.*
Area: Mathematics Education

# Awards and Scholarships

Feb-2025 **Recognition for High Performance in Teaching Practice**
*Facultad de Ciencias Exactas, Naturales y Ambientales – PUCE*

dec-2023 **Gratitude Button**
*Asociación de Profesores de la Pontificia Universidad Católica del Ecuador*

jul-2021 **UOC Scholarship for Outstanding Students from Ecuador**
*Universitat Oberta de Catalunya*

dec-2017 **Municipal Award Pedro Vicente Maldonado for the Best Published Work in the Area of Exact Sciences in 2017**
*Municipio del Distrito Metropolitano de Quito*
Work: A Primitive Associated to the Cantor Bendixson Derivative on the Real Line

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